

A Real Example for Performing Summary-Data MR Analysis with the MRMU Package

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Introduction

MRMU is an R package for Mendelian randomization analyses with one primary exposure, one outcome, and multiple potential confounder traits using genome-wide summary statistics.

This tutorial walks through a complete real-data example:

- Exposure: `Biomarker_Urate`
- Outcome: `CAD_UKB`
- Confounders: `Metabolic_SBP`, `Metabolic_DBP`
- Instrument threshold: $p \leq 5e-5$
- LD clumping: local PLINK with the 1000 Genomes EUR reference panel
- Background parameters: estimated from EUR LD scores

The full workflow has four practical steps:

1. Install and load packages.
2. Prepare and harmonise summary statistics.
3. Estimate background nuisance parameters and select clumped instruments.
4. Fit the MRMU model and read the output.

Step 0: Installation and Loading Packages

Install MRMU from GitHub:

```
# install.packages("remotes")
remotes::install_github("hxx0928/MRMU", dependencies = TRUE)
```

Load the package and a fast table-reading package:

```
library(MRMU)
library(data.table)
```

Data Download Links

This tutorial assumes the example data have been downloaded and organised in the folder structure shown below. The shared download links will be added after they are available.

data type	link
GWAS summary statistics	To be added
1000 Genomes EUR PLINK reference	To be added
EUR LD score files	To be added
macOS/Linux PLINK executable	To be added

For this tutorial, all input files are organised under one project folder:

```
bundle_dir <- "path/to/MRMU_R_github"
```

```
data_dir <- file.path(bundle_dir, "raw_gwas")
ldscore_dir <- file.path(bundle_dir, "ldscore", "eur_w_ld_chr")
bfile_prefix <- file.path(bundle_dir, "plink_reference", "all_1000G_EUR_Phase3")
plink_bin <- file.path(bundle_dir, "tools", "plink_mac", "plink")
```

Replace path/to/MRMU_R_github with the folder where you downloaded and organised the example data. The four paths mean:

object	meaning
data_dir	formatted GWAS summary-statistic files
ldscore_dir	LDSC files 1-22.12.ldscore.gz and 1-22.12.M_5_50
bfile_prefix	PLINK reference prefix without .bed/.bim/.fam suffix
plink_bin	local PLINK executable

Step 1: Prepare and Harmonise Data

1.1. Required Input Columns

Each GWAS file should already be formatted with the following columns:

column	meaning
SNP	rsID
A1	effect allele
A2	non-effect allele
Z	signed z-score

column	meaning
N	sample size
chi2	chi-square statistic
P	p-value

For example, the urate file starts like this:

```
SNP A1 A2 Z N chi2 P
rs1000000 A G -0.1127863 362754 0.01272074 0.9102
rs10000010 C T -0.5980598 362754 0.35767558 0.5498
rs10000023 T G 2.4577769 362754 6.04066718 0.01398
```

Create a file map for the four traits:

```
traits <- c("Biomarker_Urate", "Metabolic_SBP", "Metabolic_DBP", "CAD_UKB")

data.files <- data.frame(
  trait.name = traits,
  file.dir = file.path(data_dir, traits),
  stringsAsFactors = FALSE
)
```

1.2. Read LD Scores

MRMU uses SNP-level LD scores in the background model. Read and combine all 22 chromosomes:

```
ld <- rbindlist(lapply(1:22, function(chr) {
  fread(file.path(ldscore_dir, paste0(chr, ".ldscore.gz")))
}), use.names = TRUE)
```

The combined `ld` table must contain at least:

SNP, L2

1.3. Harmonise All Traits

Use `MV_harmonise()` to align all traits to the same effect allele. This function:

- merges the GWAS files by SNP;
- checks whether alleles are aligned or flipped;
- flips Z when needed;
- converts Z and N to SNP effect estimates and standard errors;
- merges in the LD score column L2.

```
dat <- MV_harmonise(data.files, ld)
```

In this example, the harmonised data contained 1,012,146 SNPs.

The output columns are:

```

SNP, A1, A2,
b_Biomarker_Urate, se_Biomarker_Urate, pval_Biomarker_Urate,
b_Metabolic_SBP, se_Metabolic_SBP, pval_Metabolic_SBP,
b_Metabolic_DBP, se_Metabolic_DBP, pval_Metabolic_DBP,
b_CAD_UKB, se_CAD_UKB, pval_CAD_UKB,
L2

```

For each trait, the effect estimate and standard error are computed as:

```

b = Z / sqrt(N)
se = 1 / sqrt(N)

```

Step 2: Estimate Background Parameters

MRMU needs two nuisance matrices for the exposure, confounders, and outcome:

matrix	dimension	interpretation
Omega	4 by 4	covariance matrix of background polygenic effects
C	4 by 4	covariance/intercept matrix capturing sample structure and correlated errors

Here the trait order is:

```

traits_for_model <- c("Biomarker_Urate", "Metabolic_SBP", "Metabolic_DBP", "CAD_UKB")

```

Estimate the LDSC-based parameters:

```

MRMU_ldsc_paras <- est_MRMU_ldsc_paras(
  data.files = data.files,
  ldscore.files = ldscore_dir
)

rownames(MRMU_ldsc_paras$Omega) <- colnames(MRMU_ldsc_paras$Omega) <- traits
rownames(MRMU_ldsc_paras$C) <- colnames(MRMU_ldsc_paras$C) <- traits
rownames(MRMU_ldsc_paras$gc) <- colnames(MRMU_ldsc_paras$gc) <- traits

diag(MRMU_ldsc_paras$C) <- pmax(diag(MRMU_ldsc_paras$C), 1)

```

The last line prevents diagonal elements of C from being smaller than 1, matching the production workflow.

For this example, the estimated genetic correlations were:

trait1	trait2	genetic correlation
Biomarker_Urate	Metabolic_SBP	0.1553
Biomarker_Urate	Metabolic_DBP	0.1316
Biomarker_Urate	CAD_UKB	0.2735
Metabolic_SBP	Metabolic_DBP	0.7748
Metabolic_SBP	CAD_UKB	0.3625
Metabolic_DBP	CAD_UKB	0.2798

The estimated Ω matrix was:

	Biomarker_Urate	Metabolic_SBP	Metabolic_DBP	CAD_UKB
Biomarker_Urate	1.163e-07	1.634e-08	1.331e-08	2.238e-08
Metabolic_SBP	1.634e-08	9.822e-08	7.293e-08	2.747e-08
Metabolic_DBP	1.331e-08	7.293e-08	9.110e-08	2.044e-08
CAD_UKB	2.238e-08	2.747e-08	2.044e-08	5.970e-08

The estimated C matrix was:

	Biomarker_Urate	Metabolic_SBP	Metabolic_DBP	CAD_UKB
Biomarker_Urate	1.1120	0.0579	0.0666	0.0529
Metabolic_SBP	0.0579	1.2026	0.8589	0.0460
Metabolic_DBP	0.0666	0.8589	1.1833	0.0163
CAD_UKB	0.0529	0.0460	0.0163	1.0296

Step 3: Select Instruments by Local LD Clumping

Use urate p-values to select candidate instruments and then LD-clump them with the local 1000 Genomes EUR reference panel.

```
p_thresh <- 5e-5
```

```
MRMUInput <- MRMU_Input(
  dat = dat,
  exposure = "Biomarker_Urate",
  confounders = c("Metabolic_SBP", "Metabolic_DBP"),
  outcome = "CAD_UKB",
  p_thresh = p_thresh,
  bfile = bfile_prefix,
  plink_bin = plink_bin,
  clump_kb = 1000,
```

```

    clump_r2 = 0.001
)

```

The clumping parameters mean:

argument	value	meaning
p_thresh	5e-5	initial IV selection p-value threshold
clump_kb	1000	remove nearby correlated SNPs within 1000 kb
clump_r2	0.001	LD threshold for pruning correlated SNPs
bfile	local 1KG EUR reference	reference genotypes for LD calculation

This run used:

PLINK v1.9.0-b.7.7 64-bit (22 Oct 2024)

The clumping summary was:

item	value
candidate urate IVs at $p \leq 5e-5$	14,090
clumped IVs	696
removed due to LD/reference absence	13,394

MRMUInput is a list. The important fields are:

field	dimension	meaning
b.exp	m by 3	SNP effects on urate, SBP, and DBP
se.exp	m by 3	standard errors for b.exp
pval.exps	m by 3	p-values for urate, SBP, and DBP
b.out	m by 1	SNP effects on CAD_UKB
se.out	m by 1	standard errors for CAD_UKB effects
L2	m by 1	SNP-level LD scores

field	dimension	meaning
Threshold	scalar	adjusted IV threshold used for selection-bias correction

Here `m` = 696 after clumping.

Step 4: Fit MRMU

Now fit the MRMU model. The order of rows and columns in `Omega` and `C` must match `c(exposure, confounders, outcome)`.

```
traits_for_model <- c(
  "Biomarker_Urate",
  "Metabolic_SBP",
  "Metabolic_DBP",
  "CAD_UKB"
)

fit.MRMU <- MRMU(
  MRdat = MRMUInput,
  exposure = "Biomarker_Urate",
  confounders = c("Metabolic_SBP", "Metabolic_DBP"),
  outcome = "CAD_UKB",
  Omega = MRMU_ldsc_paras$Omega[traits_for_model, traits_for_model],
  C = MRMU_ldsc_paras$C[traits_for_model, traits_for_model],
  Cor.SelectionBias = TRUE,
  cut.confounders = FALSE,
  tol = 1e-8,
  a = 1,
  b = 1
)
```

Important arguments:

argument	meaning
<code>Cor.SelectionBias = TRUE</code>	use the adjusted Threshold to correct IV selection bias
<code>cut.confounders = FALSE</code>	keep both SBP and DBP in this focused example
<code>tol = 1e-8</code>	convergence tolerance for the variational EM algorithm
<code>a = 1, b = 1</code>	prior hyperparameters for confounder effects

The fitted object includes:

field	meaning
beta1	estimated causal effect of urate on CAD_UKB
beta1.se	standard error
beta1.pvalue	likelihood-ratio-test p-value
nIV	number of clumped IVs
nvalid	effective number of foreground IVs
pi0	estimated foreground probability
SigmaX	fitted covariance matrix for exposure/confounder foreground effects
tau.sq	fitted variance of direct outcome effects
post	posterior quantities, including per-SNP posterior probabilities

Step 4 Result: MRMU

The LDSC-based MRMU result for this example was:

quantity	value
beta	0.01232753
standard error	0.0152602
p-value	0.41919
clumped IVs	696
effective IVs	322.2309
pi0	0.4629754
tau.sq	2.368078e-07
threshold	5e-5
method	MR-MU with LDSC Ω_{C} and local PLINK clumping

After accounting for SBP and DBP and using LDSC-estimated background parameters, the estimated effect of urate on CAD_UKB was small and not statistically significant.

Step 5: Run the No-Confounder MRAPSS Comparison

To understand how much the SBP and DBP confounders change the result, we also run a no-confounder analysis. This comparison uses the same exposure,

outcome, LD scores, local PLINK reference, and IV threshold, but does not include `Metabolic_SBP` or `Metabolic_DBP`.

This comparison is useful because it answers a concrete question:

What would the urate-CAD_UKB result look like if we ignored SBP and DBP?

5.1. Reuse the Background Parameters From Step 2

There is no need to repeat LDSC estimation for the no-confounder comparison. Step 2 already estimated Ω and C for all four traits. The no-confounder model only needs the 2 by 2 submatrices for:

```
exposure <- "Biomarker_Urate"
outcome <- "CAD_UKB"
```

Extract the relevant rows and columns from the Step 2 output:

```
C.no.conf <- MRMU_ldsc_paras$C[c(exposure, outcome), c(exposure, outcome)]
Omega.no.conf <- MRMU_ldsc_paras$Omega[c(exposure, outcome), c(exposure, outcome)]
```

For this example, the extracted matrices are:

`Omega.no.conf`

	Biomarker_Urate	CAD_UKB
Biomarker_Urate	1.163e-07	2.238e-08
CAD_UKB	2.238e-08	5.970e-08

`C.no.conf`

	Biomarker_Urate	CAD_UKB
Biomarker_Urate	1.1120	0.0529
CAD_UKB	0.0529	1.0296

Also reuse the harmonised multi-trait data from Step 1. We only keep the urate and CAD_UKB columns needed by the no-confounder model:

```
MRdat.no.conf.all <- data.frame(
  SNP = dat$SNP,
  A1 = dat$A1,
  A2 = dat$A2,
  b.exp = dat[[paste0("b_", exposure)]],
  b.out = dat[[paste0("b_", outcome)]],
  se.exp = dat[[paste0("se_", exposure)]],
  se.out = dat[[paste0("se_", outcome)]],
  pval.exp = dat[[paste0("pval_", exposure)]],
  pval.out = dat[[paste0("pval_", outcome)]],
  L2 = dat$L2
)
```

5.2. LD-Clump the Urate Instruments

Use the same IV threshold and local PLINK reference:

```
MRdat.no.conf <- MRAPSS::clump(  
  MRdat.no.conf.all,  
  IV.Threshold = p_thresh,  
  SNP_col = "SNP",  
  pval_col = "pval.exp",  
  clump_kb = 1000,  
  clump_r2 = 0.001,  
  bfile = bfile_prefix,  
  plink_bin = plink_bin  
)
```

The no-confounder clumping summary was:

item	value
candidate urate IVs at $p \leq 5e-5$	14,090
clumped IVs	696
removed due to LD/reference absence	13,394

These are the same candidate and clumped IV counts as the MRMU run because both models now reuse the same Step 1 harmonised multi-trait data.

5.3. Fit the No-Confounder Model

Fit the no-confounder model:

```
fit.no.conf <- MRAPSS::MRAPSS(  
  MRdat.no.conf,  
  exposure = exposure,  
  outcome = outcome,  
  C = C.no.conf,  
  Omega = Omega.no.conf,  
  Cor.SelectionBias = TRUE  
)
```

Extract a compact result table:

```
pvalue.no.conf <- if (!is.null(fit.no.conf$pvalue)) {  
  fit.no.conf$pvalue  
} else {  
  fit.no.conf$pval  
}  
  
res.no.conf <- data.frame(  
  pvalue.no.conf,  
  pvalue.no.conf
```

```

exposure = exposure,
outcome = outcome,
confounders = "None",
threshold = p_thresh,
beta = fit.no.conf$beta,
se = fit.no.conf$beta.se,
pvalue = pvalue.no.conf,
nIV = nrow(MRdat.no.conf),
neffIV = fit.no.conf$pi0 * nrow(MRdat.no.conf),
pi0 = fit.no.conf$pi0,
sigma_sq = fit.no.conf$sigma_sq,
tau_sq = fit.no.conf$tau_sq,
method = "MRAPSS(no confounders)"
)

```

Step 5 Result: MRAPSS Without Confounders

The no-confounder MRAPSS result was:

quantity	value
beta	0.04490893
standard error	0.01613808
p-value	5.3893e-03
clumped IVs	696
effective IVs	324.7741
pi0	0.4666295
sigma_sq	4.59646e-06
tau_sq	3.29462e-07
threshold	5e-5
method	MRAPSS without confounders

Without SBP and DBP in the model, the estimated urate effect on CAD_UKB was larger and statistically significant.

MRMU vs MRAPSS Without Confounders

The side-by-side comparison is:

method	confounders	beta	standard error	p-value	effective IVs	effective IVs	pi0
MRMU	SBP	0.01232753	0.01526020	0.41919	696	322.2309	0.4629754
	+						
	DBP						

method	confounders	beta	standard error	p-value	IVs	effective IVs	pi0
MRAPSS	none	0.044908930	0.016138085	5.3893e-03	696	324.7741	0.4666295

The no-confounder analysis gives a larger and statistically significant effect estimate. After including SBP and DBP in MRMU, the effect estimate is attenuated and no longer significant, suggesting that blood pressure traits explain part of the apparent urate-CAD_UKB association in the single-exposure analysis.

A Complete Script

The complete script is saved as:

`package/scripts/run_urate_cad_mrmu_ldsc_example.R`

Run it from the package root:

`MRMU_EXAMPLE_DIR=/path/to/MRMU_R_github Rscript scripts/run_urate_cad_mrmu_ldsc_example.R`

The script writes:

output file	content
<code>urate_cad_mrmu_ldsc_result.csv</code>	final MRMU result table
<code>urate_cad_mrmu_ldsc_fit.rds</code>	fitted model object
<code>urate_cad_mrmu_ldsc_input.rds</code>	clumped MRMU input
<code>MRMU_ldsc_paras_urate_sbp_dbp_cadukb.rds</code>	estimated Omega, C, and genetic correlations
<code>ldsc_pairwise_results.csv</code>	pairwise LDSC summary table
<code>run.log</code>	full run log

The no-confounder comparison script is saved as:

`package/scripts/run_urate_cad_mrapss_comparison.R`

Run it from the package root with:

`MRMU_EXAMPLE_DIR=/path/to/MRMU_R_github Rscript scripts/run_urate_cad_mrapss_comparison.R`

It writes:

output file	content
<code>urate_cad_mrapss_result.csv</code>	no-confounder MRAPSS result table
<code>urate_cad_mrmu_vs_mrapss_comparison.csv</code>	side-by-side MRMU vs MRAPSS comparison

output file	content
<code>urate_cad_mrapss_fit.rds</code>	fitted no-confounder model
<code>urate_cad_mrapss_input.rds</code>	clumped no-confounder input
<code>urate_cad_mrapss_ldsc_paras_from_step2.rds</code>	no-confounder Omega/C submatrices extracted from Step 2

Troubleshooting

If local clumping fails, check:

- `plink_bin` points to an executable macOS/Linux binary for your system.
- `bfile_prefix` omits the `.bed`, `.bim`, and `.fam` suffix.
- all three PLINK reference files exist in the same folder.

If LDSC parameter estimation fails, check:

- `ldscore_dir` contains all `1-22.12.ldscore.gz` files.
- `ldscore_dir` also contains all `1-22.12.M_5_50` files.
- the GWAS files contain valid `SNP`, `A1`, `A2`, `Z`, `N`, `chi2`, and `P` columns.

For large runs, save intermediate objects such as `dat`, `MRMUInput`, and `MRMU_ldsc_paras`, so the full pipeline can resume without repeating the most expensive steps.